

Online Supplement for the Following Article

Air pollution and family-related determinants of persistent wheezing and asthma onset in children – a nationwide case-control study

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Information on air pollution modeling

The DEHM/UBM system consists of a coupling of several air pollution models, covering different areas and with different resolution. These chemistry-transport models comprise the Danish Eulerian Hemispheric Model (DEHM)(1) that covers the Northern Hemisphere and includes three nested domains over Europe (50 km x 50 km), Northern Europe (16.7 km x 16.7 km) and Denmark (5.6 km x 5.6 km) for higher resolution over the area of interest (2, 3). The Urban Background Model (UBM) covers Denmark with a high resolution of 1 km x 1 km and uses background concentrations from the DEHM model as well as contributions from urban areas (4, 5). Inputs to the models include meteorological data, emissions from both anthropogenic (traffic, industry etc.) and natural emissions (e.g. uplift of sea salt from the sea surface and gases from vegetation). The models then describe the overall transport and chemical processes in the atmosphere as well as the removal processes (e.g. by rain). The model system has been developed and improved for many years and are continuously evaluated by comparison to observed concentrations from monitoring sites across Europe and show good overall performance (4). For the Danish area, more detailed evaluations have been done e.g. based on the Danish National monitoring network including both rural and urban sites (5, 6). The system has also been included in a number of exposure studies including detailed evaluations (7, 8).

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Table E1 Study population characteristics and risk factors for asthma incidence

	Cases		Controls		HR	(95% CI)
	N=122,842		N=3,069,943			
	n	(%)	n	(%)		
Basic characteristics						
Sex						
Male	64,212	(52)	1,604,416	(52)	-	
Female	58,630	(48)	1,465,527	(48)	-	
Year of birth						
Year 1997	10,320	(8)	257,713	(8)	-	
Year 1998	10,106	(8)	252,440	(8)	-	
Year 1999	9,208	(7)	230,079	(7)	-	
Year 2000	8,784	(7)	219,568	(7)	-	
Year 2001	8,299	(7)	207,389	(7)	-	
Year 2002	8,436	(7)	210,862	(7)	-	
Year 2003	8,488	(7)	212,114	(7)	-	
Year 2004	8,396	(7)	209,837	(7)	-	
Year 2005	8,170	(7)	204,213	(7)	-	
Year 2006	7,728	(6)	193,195	(6)	-	
Year 2007	7,657	(6)	191,403	(6)	-	
Year 2008	7,038	(6)	175,933	(6)	-	
Year 2009	6,113	(5)	152,798	(5)	-	
Year 2010	5,439	(4)	135,959	(4)	-	
Year 2011	4,666	(4)	116,618	(4)	-	
Year 2012	3,340	(3)	83,478	(3)	-	
Year 2013	654	(1)	16,344	(1)	-	
Age of diagnosis						
Age 1	85,507	(70)	2,137,310	(70)	-	
Age 2	15,841	(13)	395,811	(13)	-	
Age 3	6,948	(5)	173,580	(5)	-	
Age 4	3,675	(3)	91,811	(3)	-	
Age 5	2,104	(2)	52,563	(2)	-	
Age 6	1,719	(1)	42,942	(1)	-	
Age 7	1,473	(1)	36,763	(1)	-	
Age 8	1,275	(1)	31,835	(1)	-	
Age 9	1,084	(1)	27,065	(1)	-	
Age 10	968	(0.5)	24,164	(0.5)	-	
Age 11	783	(0.5)	19,555	(0.5)	-	
Age 12	575	(0.5)	14,337	(0.5)	-	
Age 13	392	(0.5)	9,787	(0.5)	-	
Age 14	304	(0.5)	7,576	(0.5)	-	
Age 15	194	(0.5)	4,844	(0.5)	-	
Year of diagnosis						
Year 1998	4,201	(3)	104,958	(3)	-	
Year 1999	6,794	(5)	169,766	(5)	-	
Year 2000	7,442	(6)	185,962	(6)	-	
Year 2001	7,795	(6)	194,780	(6)	-	
Year 2002	7,828	(6)	195,633	(6)	-	
Year 2003	7,061	(6)	176,472	(6)	-	
Year 2004	8,723	(7)	218,000	(7)	-	
Year 2005	7,946	(6)	198,571	(6)	-	
Year 2006	9,014	(7)	225,313	(7)	-	
Year 2007	8,405	(7)	210,065	(7)	-	
Year 2008	8,663	(7)	216,524	(7)	-	
Year 2009	7,735	(6)	193,347	(6)	-	
Year 2010	8,645	(7)	216,060	(7)	-	
Year 2011	7,119	(6)	177,908	(6)	-	
Year 2012	6,955	(6)	173,801	(6)	-	
Year 2013	5,564	(5)	139,018	(5)	-	
Year 2014	2,952	(3)	73,765	(3)	-	

Definition of abbreviations: N, number; HR, hazard ratio; 95% CI, 95% confidence interval.

Table E1 (continued)

	Cases		Controls		HR	(95% CI)
	N=122,842		N=3,069,943			
	n	(%)	n	(%)		
Genetic determinants						
Parental asthma						
No parents	72,798	(59)	2,191,465	(71)	1.0	(ref.)
Father	15,982	(13)	317,779	(10)	1.51	1.48 - 1.53
Mother	27,530	(23)	478,034	(16)	1.72	1.72 - 1.77
Both parents	6,532	(5)	82,665	(3)	2.40	2.34 - 2.47
Social determinants						
Education- Mother						
Primary school	22,646	(19)	456,226	(15)	1.0	(ref.)
Short education	58,953	(48)	1,428,517	(47)	0.82	0.81 - 0.83
Medium education	30,453	(25)	824,610	(27)	0.73	0.71 - 0.74
Long education	10,353	(8)	346,539	(11)	0.58	0.56 - 0.59
Education- Father						
Primary school	23,696	(20)	505,560	(17)	1.0	(ref.)
Short education	68,887	(57)	1,676,372	(55)	0.87	0.85 - 0.88
Medium education	16,512	(13)	454,841	(15)	0.76	0.75 - 0.77
Long education	12,526	(10)	400,724	(13)	0.65	0.64 - 0.67
Income - Mother*						
1 st quintile	22,542	(19)	517,744	(17)	1.0	(ref.)
2 nd quintile	43,561	(36)	1,016,642	(33)	0.96	0.95 - 0.98
3 rd quintile	32,823	(26)	831,516	(27)	0.87	0.86 - 0.89
4 th quintile	15,794	(13)	451,498	(15)	0.77	0.76 - 0.78
5 th quintile	7,454	(6)	235,749	(8)	0.69	0.67 - 0.71
Income - Father*						
1 st quintile	9,122	(7)	211,973	(7)	1.0	(ref.)
2 nd quintile	12,850	(11)	282,592	(9)	1.05	1.02 - 1.08
3 rd quintile	21,946	(18)	502,216	(17)	1.00	0.98 - 1.03
4 th quintile	33,949	(28)	830,262	(27)	0.93	0.91 - 0.95
5 th quintile	44,202	(36)	1,223,610	(40)	0.81	0.79 - 0.83
Parental education [#]						
Level 1	10,024	(8)	193,016	(6)	1.0	(ref.)
Level 2	21,869	(18)	469,738	(16)	0.89	0.87 - 0.92
Level 3	42,940	(35)	1,037,317	(34)	0.79	0.77 - 0.81
Level 4	21,404	(18)	561,113	(19)	0.73	0.71 - 0.74
Level 5	13,346	(11)	384,149	(13)	0.66	0.64 - 0.68
Level 6	6,403	(5)	201,040	(7)	0.60	0.58 - 0.62
Level 7	5,349	(5)	185,159	(6)	0.55	0.53 - 0.57
Parental income [#]						
Level 1	4,769	(4)	102,580	(3)	1.0	(ref.)
Level 2	6,589	(5)	141,363	(5)	0.99	0.95 - 1.03
Level 3	11,127	(9)	242,498	(8)	0.96	0.94 - 1.00
Level 4	16,838	(14)	382,263	(12)	0.93	0.89 - 0.96
Level 5	23,713	(20)	576,023	(19)	0.85	0.83 - 0.88
Level 6	25,970	(21)	663,495	(22)	0.80	0.78 - 0.83
Level 7	18,761	(15)	504,710	(17)	0.75	0.73 - 0.78
Level 8	9,329	(8)	276,324	(9)	0.68	0.66 - 0.71
Level 9	4,963	(4)	161,047	(5)	0.62	0.60 - 0.65
Mother smoking						
Non-smoker	83,840	(79)	2,219,875	(84)	1.0	(ref.)
Quit smoking in 1 st trimester	2,439	(2)	48,333	(2)	1.24	1.19 - 1.29
Quit smoking after the 1 st trimester	587	(1)	11,682	(1)	1.28	1.17 - 1.38
Smokes up to 5 cigarettes daily	5,782	(5)	114,287	(4)	1.30	1.26 - 1.33
Smokes 6 to 10 cigarettes daily	7,215	(7)	137,076	(5)	1.38	1.35 - 1.42
Smokes 11 to 20 cigarettes daily	5,470	(5)	100,182	(3)	1.43	1.39 - 1.46
Smokes more than 20 cigarettes daily	681	(1)	13,134	(1)	1.42	1.32 - 1.53

Definition of abbreviations: n, number; HR, hazard ratio; 95% CI, 95% confidence interval; SES, socio-economic status; ref., reference.

Description of the Danish education system: Primary school: primary and secondary education. Short education: 2-3 years (Without upper secondary education). Medium education: bachelor's programme, professional bachelor's programme, academy profession programme (requires completion of upper secondary education). Long education: master's programmes (candidatus) PhD.

*Quintiles are based on the income of the general population.

[#] Parental income and parental education were generated adding the level of income (level 1-5) or education (level 1-4) for each parent generating a 7-level categorical variable for parental education and a 9-level categorical variable for parental income.

Table E2 Risk of asthma and persistent wheezing with mutual adjustment for family-related factors.

	Crude model		Adjusted model	
	HR	(95% CI)	HR	(95% CI)
Genetic determinants				
Parental asthma				
No parents	1.0 (ref.)		1.0 (ref.)	
Father	1.51	1.48 - 1.53	1.49	(1.47 - 1.52)
Mother	1.72	1.72 - 1.77	1.67	(1.64 - 1.69)
Both parents	2.40	2.34 - 2.47	2.29	(2.22 - 2.35)
Social determinants				
Parental education [#]				
Level 1	1.0 (ref.)		1.0 (ref.)	
Level 2	0.89	0.87 - 0.92	0.98	(0.95 - 1.00)
Level 3	0.79	0.77 - 0.81	0.93	(0.90 - 0.95)
Level 4	0.73	0.71 - 0.74	0.88	(0.86 - 0.91)
Level 5	0.66	0.64 - 0.68	0.82	(0.80 - 0.85)
Level 6	0.60	0.58 - 0.62	0.76	(0.73 - 0.79)
Level 7	0.55	0.53 - 0.57	0.72	(0.69 - 0.75)
Parental income [#]				
Level 1	1.0 (ref.)		1.0 (ref.)	
Level 2	0.99	0.95 - 1.03	1.04	(1.00 - 1.08)
Level 3	0.96	0.94 - 1.00	1.03	(0.99 - 1.07)
Level 4	0.93	0.89 - 0.96	1.03	(0.99 - 1.06)
Level 5	0.85	0.83 - 0.88	0.99	(0.96 - 1.03)
Level 6	0.80	0.78 - 0.83	0.95	(0.92 - 0.98)
Level 7	0.75	0.73 - 0.78	0.93	(0.89 - 0.96)
Level 8	0.68	0.66 - 0.71	0.88	(0.84 - 0.92)
Level 9	0.62	0.60 - 0.65	0.85	(0.81 - 0.89)
Mother's smoking behaviour during				
Non-smoker	1.0 (ref.)		1.0 (ref.)	
Quit smoking during pregnancy	1.33	1.28 - 1.39	1.23	(1.18 - 1.28)
Smoking during pregnancy	1.39	1.37 - 1.42	1.20	(1.18 - 1.22)

Abbreviations: HR, hazard ratio; 95% CI, 95% confidence interval; ref., reference.

Model description: Conditional logistic regression analyses of the adjusted effect of determinants on asthma. Adjustment for parental asthma, parental education, parental income, and maternal smoking during pregnancy.

[#]Parental income and parental education were generated adding the level of income (level 1-5) or education (level 1-4) for each parent generating a 7-level categorical variable for parental education and a 9-level variable for parental income.

Table E3 Risk of asthma and persistent wheezing with mutual adjustment for the determinants over the study period.

	1998-2002		2003-2008		2009-2014	
	HR	(95% CI)	HR	(95% CI)	HR	(95% CI)
Genetic determinants						
Parental asthma						
No parents	1.0 (ref.)		1.0 (ref.)		1.0 (ref.)	
Father	1.47	(1.41 - 1.54)	1.50	(1.46 - 1.55)	1.50	(1.45 - 1.55)
Mother	1.62	(1.57 - 1.68)	1.68	(1.64 - 1.72)	1.67	(1.63 - 1.72)
Both parents	2.18	(2.05 - 2.31)	2.34	(2.24 - 2.45)	2.29	(2.18 - 2.42)
Social determinants						
Parental education [#]						
Level 1	1.0 (ref.)		1.0 (ref.)		1.0 (ref.)	
Level 2	0.97	(0.92 - 1.03)	0.97	(0.93 - 1.01)	0.96	(0.92 - 1.01)
Level 3	0.96	(0.91 - 1.01)	0.91	(0.87 - 0.95)	0.93	(0.88 - 0.97)
Level 4	0.91	(0.85 - 0.96)	0.85	(0.82 - 0.89)	0.89	(0.85 - 0.94)
Level 5	0.89	(0.83 - 0.95)	0.79	(0.75 - 0.83)	0.83	(0.78 - 0.88)
Level 6	0.76	(0.70 - 0.83)	0.76	(0.71 - 0.80)	0.76	(0.71 - 0.81)
Level 7	0.72	(0.70 - 1.08)	0.68	(0.64 - 0.72)	0.75	(0.70 - 0.81)
Parental income [#]						
Level 1	1.0 (ref.)		1.0 (ref.)		1.0 (ref.)	
Level 2	0.94	(0.74 - 1.18)	1.08	(1.01 - 1.16)	1.04	(0.99 - 1.10)
Level 3	1.05	(0.84 - 1.30)	1.10	(1.03 - 1.17)	1.00	(0.95 - 1.05)
Level 4	1.05	(0.86 - 1.30)	1.10	(1.04 - 1.17)	0.98	(0.93 - 1.02)
Level 5	1.06	(0.86 - 1.31)	1.07	(1.01 - 1.14)	0.92	(0.88 - 0.97)
Level 6	1.00	(0.81 - 1.23)	1.02	(0.96 - 1.08)	0.90	(0.85 - 0.95)
Level 7	0.99	(0.80 - 1.22)	0.99	(0.93 - 1.05)	0.86	(0.81 - 0.91)
Level 8	0.93	(0.75 - 1.15)	0.92	(0.86 - 0.98)	0.85	(0.79 - 0.91)
Level 9	0.87	(0.70 - 1.08)	0.92	(0.84 - 0.99)	0.83	(0.76 - 0.89)
Mother's smoking behaviour during						
Non-smoker	1.0 (ref.)		1.0 (ref.)		1.0 (ref.)	
Quit smoking during pregnancy	1.32	(1.21 - 1.44)	1.25	(1.18 - 1.32)	1.16	(1.09 - 1.23)
Smoking during pregnancy	1.27	(1.23 - 1.32)	1.19	(1.16 - 1.22)	1.14	(1.11 - 1.19)

Abbreviations: HR, hazard ratio; 95% CI, 95% confidence interval; ref., reference.

Model description: Conditional logistic regression analyses of the adjusted effect of determinants on asthma by years. Adjustment for parental

[#]Parental income and parental education were generated adding the level of income (level 1-5) or education (level 1-4) for each parent generating a 7-level categorical variable for parental education and a 9-level variable for parental income.

Table E4 Correlations between air pollutants

Air pollutants (µg/m ³)	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
(1) O ₃	-												
(2) NO ₂	-0.62	-											
(3) PM _{2.5}	-0.53	0.47	-										
(4) PM ₁₀	-0.49	0.32	0.95	-									
(5) NO _x	-0.61	0.98	0.48	0.34	-								
(6) SO ₂	-0.61	0.64	0.72	0.59	0.65	-							
(7) SO ₄ ⁻	-0.14	0.39	0.71	0.63	0.39	0.66	-						
(8) NO ₃	-0.23	0.23	0.82	0.76	0.22	0.52	0.71	-					
(9) NH ₄ ⁺	-0.30	0.26	0.88	0.83	0.25	0.61	0.83	0.92	-				
(10) SIA	-0.21	0.33	0.83	0.75	0.32	0.63	0.92	0.92	0.95	-			
(11) EC	-0.69	0.70	0.62	0.47	0.69	0.81	0.33	0.29	0.39	0.34	-		
(12) OC	0.08	-0.49	0.02	0.38	0.71	0.72	0.25	0.23	0.29	0.26	0.98	-	
(13) Seasalt	0.08	-0.49	0.02	0.29	-0.45	-0.25	-0.14	-0.06	-0.01	-0.10	-0.32	-0.37	-

Pearsons correlation coefficient (r) describes correlations between air pollutants. All correlations were statistical significant with a p-value <0.05

Table E5 Effects of air pollution on the risk of asthma and persistent wheezing.

Air pollutants	Increase	Model 1	Model 2	Model 3
		HR (95% CI)	HR (95% CI)	HR (95% CI)
O ₃	10 µg/m ³	1.01 (0.99 - 1.02)	1.01 (0.99 - 1.02)	0.96 (0.95 - 0.97)
NO ₂	10 µg/m ³	0.98 (0.97 - 0.99)	0.98 (0.97 - 0.99)	1.04 (1.03 - 1.04)
NO _x	10 µg/m ³	1.00 (0.99 - 1.00)	0.99 (0.98 - 1.00)	1.04 (1.03 - 1.04)
SO ₂	5 µg/m ³	0.98 (0.96 - 0.99)	0.97 (0.96 - 0.99)	1.02 (1.01 - 1.04)
PM _{2.5}	5 µg/m ³	1.04 (1.02 - 1.05)	1.03 (1.01 - 1.05)	1.05 (1.03 - 1.07)
PM ₁₀	5 µg/m ³	1.06 (1.03 - 1.08)	1.05 (1.02 - 1.07)	1.04 (1.02 - 1.06)
SIA	5 µg/m ³	1.03 (1.02 - 1.05)	1.02 (1.01 - 1.04)	1.01 (0.99 - 1.02)
SO ₄ ⁻	5 µg/m ³	0.99 (0.96 - 1.03)	0.98 (0.96 - 1.02)	1.00 (0.97 - 1.03)
NO ₃	5 µg/m ³	1.07 (1.05 - 1.09)	1.06 (1.04 - 1.08)	1.03 (1.02 - 1.05)
NH ₄ ⁺	1 µg/m ³	1.21 (1.14 - 1.28)	1.17 (1.10 - 1.24)	1.03 (0.97 - 1.10)
EC	1 µg/m ³	0.80 (0.76 - 0.84)	0.78 (0.74 - 0.82)	1.05 (0.99 - 1.10)
OC	1 µg/m ³	0.88 (0.86 - 0.91)	0.87 (0.85 - 0.90)	1.01 (0.99 - 1.05)
Seasalt	1 µg/m ³	0.99 (0.99 - 1.01)	1.00 (0.99 - 1.01)	0.97 (0.96 - 0.97)

Abbreviations: HR, hazard ratio; 95% CI, 95% confidence interval; IQR, interquartile range

Model description: Model 1: sex, age, calendar year. Model 2: Model 1+ parental asthma. Model 3: Model 2 + parental income + parental education.

Table E6 Effects of air pollution on the risk of asthma and persistent wheezing by exposure-time windows.

Air pollutants	Increase	Model 3		
		HR (95% CI)	HR (95% CI)	HR (95% CI)
		3 months	6 months	12 months
O ₃	10 µg/m ³	0.97 (0.96 - 0.98)	0.97 (0.96 - 0.97)	0.96 (0.95 - 0.97)
NO ₂	10 µg/m ³	1.03 (1.02 - 1.04)	1.03 (1.02 - 1.04)	1.04 (1.03 - 1.04)
NO _x	10 µg/m ³	1.03 (1.02 - 1.04)	1.03 (1.02 - 1.04)	1.04 (1.03 - 1.04)
SO ₂	5 µg/m ³	1.04 (1.03 - 1.06)	1.06 (1.04 - 1.08)	1.02 (1.01 - 1.04)
PM _{2.5}	5 µg/m ³	1.03 (1.02 - 1.05)	1.02 (1.01 - 1.04)	1.05 (1.03 - 1.07)
PM ₁₀	5 µg/m ³	0.99 (0.97 - 1.01)	1.03 (1.01 - 1.05)	1.04 (1.02 - 1.06)
SIA	5 µg/m ³	1.01 (0.99 - 1.03)	1.01 (0.99 - 1.03)	1.01 (0.99 - 1.02)
SO ₄ ⁻	5 µg/m ³	1.00 (0.97 - 1.03)	0.99 (0.96 - 1.02)	1.00 (0.97 - 1.03)
NO ₃	5 µg/m ³	1.01 (0.99 - 1.02)	1.02 (1.01 - 1.04)	1.03 (1.02 - 1.05)
NH ₄ ⁺	1 µg/m ³	1.00 (0.95 - 1.05)	0.97 (0.91 - 1.02)	1.03 (0.97 - 1.10)
EC	1 µg/m ³	0.99 (0.95 - 1.05)	1.00 (0.95 - 1.06)	1.05 (0.99 - 1.10)
OC	1 µg/m ³	0.98 (0.96 - 1.01)	0.99 (0.96 - 1.02)	1.01 (0.99 - 1.05)
Seasalt	1 µg/m ³	0.97 (0.97 - 0.98)	0.97 (0.96 - 0.98)	0.97 (0.96 - 0.97)

Abbreviations: HR, hazard ratio; 95% CI, 95% confidence interval; IQR, interquartile range

Model description: adjustment for sex, age, calendar year, parental asthma, parental income, and parental education.

Table E7 Effects of air pollution on the risk of asthma and persistent wheezing by IQR or $\mu\text{g}/\text{m}^3$.

Air pollutants	$\mu\text{g}/\text{m}^3$	IQR
	HR (95% CI)	HR (95% CI)
	12 months	12 months
O ₃	0.96 (0.95 - 0.97)	0.94 (0.92 - 0.95)
NO ₂	1.04 (1.03 - 1.04)	1.01 (0.99 - 1.03)
NO _x	1.04 (1.03 - 1.04)	1.02 (0.99 - 1.04)
SO ₂	1.02 (1.01 - 1.04)	1.06 (1.03 - 1.08)
PM _{2.5}	1.05 (1.03 - 1.07)	1.08 (1.05 - 1.10)
PM ₁₀	1.04 (1.02 - 1.06)	1.01 (0.98 - 1.03)
SIA	1.01 (0.99 - 1.02)	1.01 (0.98 - 1.04)
SO ₄ ⁻	1.00 (0.97 - 1.03)	1.02 (0.98 - 1.07)
NO ₃	1.03 (1.02 - 1.05)	1.03 (1.02 - 1.06)
NH ₄ ⁺	1.03 (0.97 - 1.10)	1.00 (0.97 - 1.03)
EC	1.05 (0.99 - 1.10)	0.97 (0.96 - 0.99)
OC	1.01 (0.99 - 1.05)	0.97 (0.95 - 0.98)
Seasalt	0.97 (0.96 - 0.97)	0.96 (0.94 - 0.98)

Abbreviations: HR, hazard ratio; 95% CI, 95% confidence interval; IQR, interquartile range

Model description: adjusted for sex, age, calendar year, parental asthma, parental income, and parental education.

Table E8 Effects of air pollution on the risk of asthma and persistent wheezing with adjustment for each pollutant at the time (two-pollutant models)

Air pollutants	Increase	O ₃	NO ₂	PM _{2.5}	PM ₁₀
		HR (95% CI)	HR (95% CI)	HR (95% CI)	HR (95% CI)
		12 months	12 months	12 months	12 months
O ₃	1 µg/m ³	-	1.02 (1.01 - 1.02)	1.02 (1.01 - 1.03)	1.01 (1.01 - 1.02)
NO ₂	1 µg/m ³	1.02 (1.01 - 1.02)	-	0.96 (0.94 - 0.99)	1.01 (1.01 - 1.02)
NO _x	1 µg/m ³	1.01 (1.01 - 1.02)	0.99 (0.98 - 1.01)	1.01 (1.01 - 1.02)	1.01 (1.01 - 1.02)
SO ₂	1 µg/m ³	0.99 (0.98 - 0.99)	1.01 (1.01 - 1.02)	1.02 (1.01 - 1.03)	1.01 (1.01 - 1.02)
PM _{2.5}	1 µg/m ³	0.99 (0.98 - 0.99)	1.01 (1.01 - 1.02)	-	0.99 (0.98 - 0.99)
PM ₁₀	1 µg/m ³	0.99 (0.98 - 0.99)	1.01 (1.01 - 1.02)	1.04 (1.03 - 1.05)	-
SIA	1 µg/m ³	0.99 (0.98 - 0.99)	1.01 (1.01 - 1.02)	1.04 (1.03 - 1.05)	1.01 (1.01 - 1.02)
SO ₄ ⁻	1 µg/m ³	0.99 (0.98 - 0.99)	1.01 (1.01 - 1.02)	1.03 (1.03 - 1.04)	1.01 (1.01 - 1.02)
NO ₃	1 µg/m ³	0.99 (0.98 - 0.99)	1.01 (1.01 - 1.02)	1.04 (1.03 - 1.05)	1.01 (1.01 - 1.03)
NH ₄ ⁺	1 µg/m ³	0.99 (0.98 - 0.99)	1.01 (1.01 - 1.02)	1.04 (1.03 - 1.05)	1.01 (1.01 - 1.03)
EC	1 µg/m ³	0.99 (0.98 - 0.99)	1.02 (1.01 - 1.02)	1.02 (1.02 - 1.03)	1.02 (1.01 - 1.02)
OC	1 µg/m ³	0.98 (0.98 - 0.99)	1.02 (1.01 - 1.02)	1.03 (1.02 - 1.03)	1.02 (1.01 - 1.02)
Seasalt	1 µg/m ³	0.99 (0.98 - 0.99)	1.02 (1.00 - 1.01)	1.02 (1.01 - 1.02)	1.03 (1.02 - 1.03)

Air pollutants	Increase	NO _x	SO ₂	SO ₄	NO ₃
		HR (95% CI)	HR (95% CI)	HR (95% CI)	HR (95% CI)
		12 months	12 months	12 months	12 months
O ₃	1 µg/m ³	1.00 (1.00 - 1.01)	1.01 (1.00 - 1.01)	1.00 (0.99 - 1.02)	1.00 (1.00 - 1.01)
NO ₂	1 µg/m ³	1.01 (1.00 - 1.01)	1.03 (1.01 - 1.06)	1.00 (0.98 - 1.01)	1.01 (1.00 - 1.01)
NO _x	1 µg/m ³	-	1.08 (1.05 - 1.11)	0.99 (0.98 - 1.01)	1.01 (1.00 - 1.01)
SO ₂	1 µg/m ³	1.00 (1.00 - 1.01)	-	0.98 (0.96 - 0.99)	1.01 (0.99 - 1.01)
PM _{2.5}	1 µg/m ³	1.01 (1.01 - 1.02)	1.01 (0.99 - 1.01)	0.98 (0.96 - 0.99)	0.98 (0.97 - 0.99)
PM ₁₀	1 µg/m ³	1.01 (1.01 - 1.02)	1.02 (1.01 - 1.02)	1.03 (1.01 - 1.06)	0.99 (0.98 - 1.01)
SIA	1 µg/m ³	1.00 (1.00 - 1.01)	1.02 (1.01 - 1.02)	1.03 (1.00 - 1.06)	0.96 (0.93 - 0.99)
SO ₄ ⁻	1 µg/m ³	0.99 (0.98 - 1.01)	1.02 (1.01 - 1.03)	-	0.99 (0.98 - 1.00)
NO ₃	1 µg/m ³	1.00 (1.00 - 1.01)	1.02 (1.01 - 1.02)	1.03 (1.01 - 1.05)	-
NH ₄ ⁺	1 µg/m ³	1.00 (1.00 - 1.01)	1.02 (1.01 - 1.02)	1.03 (1.00 - 1.05)	1.01 (0.99 - 1.03)
EC	1 µg/m ³	1.00 (1.00 - 1.01)	1.04 (1.03 - 1.05)	1.02 (1.00 - 1.04)	1.01 (0.99 - 1.01)
OC	1 µg/m ³	1.00 (1.00 - 1.01)	1.04 (1.03 - 1.04)	1.02 (1.00 - 1.04)	1.01 (0.99 - 1.01)
Seasalt	1 µg/m ³	1.00 (1.00 - 1.01)	1.01 (1.00 - 1.02)	0.99 (0.98 - 1.01)	1.00 (0.99 - 1.01)

Air pollutants	Increase	NH ₄ ⁺	SIA	EC	OC
		HR (95% CI)	HR (95% CI)	HR (95% CI)	HR (95% CI)
		12 months	12 months	12 months	12 months
O ₃	1 µg/m ³	1.06 (0.99 - 1.12)	1.00 (1.00 - 1.01)	0.76 (0.70 - 0.82)	0.84 (0.81 - 0.88)
NO ₂	1 µg/m ³	1.06 (1.00 - 1.13)	1.00 (0.99 - 1.01)	0.61 (0.56 - 0.67)	0.76 (0.72 - 0.80)
PM _{2.5}	1 µg/m ³	0.84 (0.78 - 0.91)	0.98 (0.98 - 0.99)	0.98 (0.93 - 1.04)	0.98 (0.95 - 1.01)
PM ₁₀	1 µg/m ³	0.94 (0.87 - 1.01)	0.99 (0.99 - 1.00)	1.07 (1.02 - 1.12)	1.03 (0.99 - 1.05)
NO _x	1 µg/m ³	1.07 (1.01 - 1.14)	1.00 (0.99 - 1.01)	0.68 (0.63 - 0.74)	0.81 (0.77 - 0.84)
SO ₂	1 µg/m ³	1.04 (0.98 - 1.11)	1.00 (0.99 - 1.01)	0.73 (0.67 - 0.80)	0.85 (0.82 - 0.89)
SO ₄ ⁻	1 µg/m ³	0.97 (0.89 - 1.04)	0.99 (0.98 - 1.00)	1.02 (0.96 - 1.08)	1.00 (0.97 - 1.03)
NO ₃	1 µg/m ³	0.92 (0.77 - 1.09)	1.03 (1.01 - 1.05)	1.06 (1.00 - 1.11)	1.02 (0.99 - 1.05)
NH ₄ ⁺	1 µg/m ³	-	1.01 (1.00 - 1.02)	1.05 (1.00 - 1.12)	1.02 (0.99 - 1.05)
SIA	1 µg/m ³	0.88 (0.76 - 1.01)	-	1.00 (1.00 - 1.01)	1.02 (0.99 - 1.05)
EC	1 µg/m ³	1.04 (0.98 - 1.11)	1.00 (1.00 - 1.01)	-	0.40 (0.30 - 0.53)
OC	1 µg/m ³	1.04 (0.98 - 1.11)	1.00 (1.00 - 1.01)	6.01 (3.49 - 10.4)	-
Seasalt	1 µg/m ³	1.02 (0.96 - 1.08)	1.00 (0.99 - 1.01)	0.86 (0.80 - 0.92)	0.91 (0.88 - 0.95)

Air pollutants	Increase	Seasalt
		HR (95% CI)
		12 months
O ₃	1 µg/m ³	0.98 (0.97 - 0.99)
NO ₂	1 µg/m ³	0.99 (0.98 - 1.01)
PM _{2.5}	1 µg/m ³	0.97 (0.97 - 0.98)
PM ₁₀	1 µg/m ³	0.96 (0.95 - 0.96)
NO _x	1 µg/m ³	0.99 (0.98 - 1.01)
SO ₂	1 µg/m ³	0.98 (0.97 - 0.99)
SO ₄ ⁻	1 µg/m ³	0.97 (0.96 - 0.97)
NO ₃	1 µg/m ³	0.97 (0.96 - 0.97)
NH ₄ ⁺	1 µg/m ³	0.97 (0.96 - 0.97)
SIA	1 µg/m ³	0.97 (0.96 - 0.97)
EC	1 µg/m ³	0.96 (0.94 - 0.97)
OC	1 µg/m ³	0.96 (0.95 - 0.97)
Seasalt	1 µg/m ³	-

Abbreviations: HR, hazard ratio; 95% CI, 95% confidence interval.

Model description: Conditional logistic regression analyses of the effect of average 12 months air pollutant concentrations on asthma. Adjustment for age, calendar year, parental asthma, parental education, parental income, and for each pollutant at the time.

Table E9 Adjusted effects of air pollution on the risk of asthma and persistent wheezing with additional adjustment for the mother's smoking during pregnancy

Air pollutants	Increase	HR (95% CI)	HR (95% CI)	HR (95% CI)
		3 months	6 months	12 months
O ₃	10 µg/m ³	0.97 (0.96 - 0.98)	0.97 (0.96 - 0.98)	0.97 (0.95 - 0.98)
NO ₂	10 µg/m ³	1.02 (1.01 - 1.04)	1.03 (1.01 - 1.04)	1.03 (1.02 - 1.04)
NO _x	10 µg/m ³	1.02 (1.02 - 1.03)	1.02 (1.02 - 1.04)	1.03 (1.02 - 1.04)
SO ₂	5 µg/m ³	1.03 (1.01 - 1.05)	1.04 (1.03 - 1.06)	1.01 (0.94 - 1.03)
PM _{2.5}	5 µg/m ³	1.02 (1.01 - 1.04)	1.01 (0.99 - 1.03)	1.05 (1.03 - 1.07)
PM ₁₀	5 µg/m ³	0.98 (0.97 - 1.01)	1.03 (1.01 - 1.05)	1.04 (1.01 - 1.06)
SIA	5 µg/m ³	1.01 (0.99 - 1.03)	1.00 (0.98 - 1.02)	1.00 (0.98 - 1.02)
SO ₄ ⁻	5 µg/m ³	0.99 (0.96 - 1.02)	0.98 (0.95 - 1.01)	0.99 (0.97 - 1.03)
NO ₃	5 µg/m ³	1.01 (0.99 - 1.03)	1.02 (1.00 - 1.04)	1.03 (1.00 - 1.05)
NH ₄ ⁺	1 µg/m ³	0.99 (0.94 - 1.05)	0.96 (0.90 - 1.03)	1.04 (0.97 - 1.11)
EC	1 µg/m ³	0.93 (0.88 - 0.97)	0.94 (0.89 - 0.99)	1.01 (0.95 - 1.07)
OC	1 µg/m ³	0.95 (0.92 - 0.97)	0.96 (0.93 - 0.98)	0.99 (0.97 - 1.02)
Seasalt	1 µg/m ³	0.97 (0.97 - 0.98)	0.98 (0.97 - 0.99)	0.97 (0.96 - 0.98)

Abbreviations: HR, hazard ratio; 95% CI, 95% confidence interval.

Model description: Conditional logistic regression analyses of the effect of average 12 months air pollutant concentrations on asthma. Model 3 + adjustment for the maternal smoking during pregnancy.

Table E10 Effects of air pollution on the risk of asthma and persistent wheezing over the study period

Air pollutants	Increase	1998-2002			2003-2008			2009-2014		
		HR (95% CI)	HR (95% CI)	HR (95% CI)	HR (95% CI)	HR (95% CI)	HR (95% CI)	HR (95% CI)	HR (95% CI)	HR (95% CI)
		3 months	6 months	12 months	3 months	6 months	12 months	3 months	6 months	12 months
O ₃	10 µg/m ³	0.93 (0.91 - 0.94)	0.92 (0.91 - 0.94)	0.92 (0.90 - 0.94)	0.98 (0.96 - 0.99)	0.98 (0.96 - 0.99)	0.98 (0.97 - 1.00)	0.99 (0.97 - 1.01)	0.99 (0.97 - 1.01)	0.98 (0.96 - 1.01)
NO ₂	10 µg/m ³	1.09 (1.07 - 1.11)	1.10 (1.08 - 1.12)	1.10 (1.08 - 1.12)	1.01 (0.99 - 1.03)	1.01 (0.99 - 1.03)	1.03 (1.01 - 1.04)	0.97 (0.94 - 0.99)	0.97 (0.94 - 0.99)	0.96 (0.94 - 0.99)
NO _x	10 µg/m ³	1.07 (1.05 - 1.08)	1.06 (1.05 - 1.08)	1.08 (1.06 - 1.09)	1.01 (1.00 - 1.02)	1.02 (1.00 - 1.03)	1.02 (1.01 - 1.04)	0.98 (0.96 - 1.01)	0.98 (0.96 - 1.00)	0.99 (0.97 - 1.01)
SO ₂	5 µg/m ³	1.12 (1.09 - 1.15)	1.16 (1.14 - 1.20)	1.12 (1.09 - 1.16)	1.03 (1.01 - 1.05)	1.03 (1.02 - 1.07)	1.01 (0.98 - 1.03)	0.97 (0.94 - 1.01)	0.97 (0.94 - 1.01)	0.98 (0.96 - 1.01)
PM _{2.5}	5 µg/m ³	1.05 (1.02 - 1.08)	1.04 (1.01 - 1.07)	1.04 (1.01 - 1.07)	1.02 (1.02 - 1.08)	1.01 (0.98 - 1.04)	1.05 (1.03 - 1.08)	1.04 (1.01 - 1.08)	1.00 (0.95 - 1.03)	1.08 (1.03 - 1.14)
PM ₁₀	5 µg/m ³	0.96 (0.93 - 0.99)	1.01 (0.97 - 1.05)	1.02 (0.98 - 1.07)	0.99 (0.96 - 1.02)	1.04 (1.01 - 1.07)	0.99 (0.95 - 1.03)	1.02 (0.99 - 1.06)	1.04 (1.01 - 1.08)	1.07 (1.04 - 1.11)
SIA	5 µg/m ³	1.03 (1.01 - 1.05)	1.02 (0.99 - 1.04)	1.03 (1.00 - 1.06)	0.99 (0.97 - 1.01)	0.98 (0.96 - 1.01)	1.00 (0.97 - 1.02)	1.04 (1.01 - 1.06)	1.03 (1.00 - 1.06)	0.98 (0.95 - 1.02)
SO ₄ ⁻	5 µg/m ³	1.06 (1.01 - 1.11)	1.00 (0.95 - 1.06)	0.99 (0.94 - 1.05)	0.97 (0.94 - 1.01)	0.99 (0.95 - 1.03)	1.01 (0.97 - 1.04)	0.98 (0.91 - 1.06)	0.81 (0.11 - 5.93)	0.01 (0.97 - 1.03)
NO ₃	5 µg/m ³	1.01 (0.98 - 1.04)	1.03 (0.99 - 1.06)	1.07 (1.04 - 1.09)	0.99 (0.97 - 1.02)	0.99 (0.96 - 1.03)	1.01 (0.98 - 1.04)	1.02 (0.99 - 1.05)	1.03 (1.00 - 1.07)	0.93 (0.87 - 0.99)
NH ₄ ⁺	1 µg/m ³	0.92 (0.85 - 1.00)	0.86 (0.79 - 0.95)	0.97 (0.88 - 1.07)	0.98 (0.91 - 1.07)	0.97 (0.89 - 1.06)	1.00 (0.92 - 1.09)	1.19 (1.06 - 1.33)	1.17 (1.03 - 1.33)	1.26 (1.09 - 1.45)
EC	1 µg/m ³	1.54 (1.42 - 1.67)	1.59 (1.45 - 1.74)	1.69 (1.54 - 1.86)	0.89 (0.82 - 0.96)	0.89 (0.82 - 0.96)	0.93 (0.86 - 1.01)	0.71 (0.64 - 0.78)	0.70 (0.63 - 0.78)	0.74 (0.66 - 0.82)
OC	1 µg/m ³	1.21 (1.15 - 1.26)	1.24 (1.18 - 1.30)	1.29 (1.22 - 1.35)	0.93 (0.89 - 0.96)	0.93 (0.89 - 0.97)	0.96 (0.92 - 1.00)	0.83 (0.79 - 0.88)	0.84 (0.79 - 0.89)	0.86 (0.82 - 0.91)
Seasalt	1 µg/m ³	0.93 (0.92 - 0.94)	0.92 (0.91 - 0.94)	0.91 (0.90 - 0.93)	0.98 (0.97 - 0.99)	0.98 (0.97 - 0.99)	0.98 (0.97 - 0.99)	1.00 (0.99 - 1.02)	1.00 (0.99 - 1.01)	1.00 (0.99 - 1.02)

Abbreviations: HR, hazard ratio; 95% CI, 95% confidence interval; IQR, interquartile range

Model description: adjustment for sex, age, calendar year, parental asthma, parental income, parental education.

Table E11 Adjusted effects of air pollution on the risk of asthma and persistent wheezing in firstborns.

Air pollutants	Increase	First-born children
		HR (95% CI)
O ₃	10 µg/m ³	0.91 (0.89 - 0.92)
NO ₂	10 µg/m ³	1.11 (1.09 - 1.12)
NO _x	10 µg/m ³	1.08 (1.07 - 1.09)
SO ₂	5 µg/m ³	1.09 (1.07 - 1.12)
PM _{2.5}	5 µg/m ³	1.08 (1.05 - 1.11)
PM ₁₀	5 µg/m ³	1.02 (0.99 - 1.05)
SIA	5 µg/m ³	1.00 (0.98 - 1.02)
SO ₄ ⁻	5 µg/m ³	1.03 (0.99 - 1.08)
NO ₃	5 µg/m ³	1.02 (0.99 - 1.05)
NH ₄ ⁺	1 µg/m ³	0.93 (0.86 - 1.02)
EC	1 µg/m ³	1.49 (1.39 - 1.60)
OC	1 µg/m ³	1.22 (1.17 - 1.26)
Seasalt	1 µg/m ³	0.94 (0.93 - 0.95)

Abbreviations: HR, hazard ratio; 95% CI, 95% confidence interval.

Model description: Conditional logistic regression analyses of the effect of average 12 months air pollutant concentrations on asthma. Adjustment for age, calendar year, parental asthma, parental education and parental income.

Table E12 Adjusted effects of air pollution on the risk of asthma and persistent wheezing by sex.

Air pollutants	Increase	Sex	
		HR (95% CI)	HR (95% CI)
		Girls	Boys
O ₃	10 µg/m ³	0.96 (0.95 - 0.98)	0.97 (0.95 - 0.98)
NO ₂	10 µg/m ³	1.04 (1.03 - 1.06)	1.03 (1.01 - 1.05)
NO _x	10 µg/m ³	1.04 (1.02 - 1.05)	1.03 (1.02 - 1.05)
SO ₂	5 µg/m ³	1.03 (1.02 - 1.06)	1.01 (0.99 - 1.03)
PM _{2.5}	5 µg/m ³	1.07 (1.04 - 1.09)	1.03 (1.01 - 1.06)
PM ₁₀	5 µg/m ³	1.04 (1.01 - 1.08)	1.03 (1.00 - 1.06)
SIA	5 µg/m ³	1.01 (0.99 - 1.03)	1.00 (0.98 - 1.03)
SO ₄ ⁻	5 µg/m ³	1.00 (0.96 - 1.05)	1.00 (0.96 - 1.04)
NO ₃	5 µg/m ³	1.03 (1.00 - 1.06)	1.04 (1.01 - 1.06)
NH ₄ ⁺	1 µg/m ³	1.03 (0.94 - 1.12)	1.03 (0.95 - 1.12)
EC	1 µg/m ³	1.06 (0.99 - 1.15)	1.04 (0.97 - 1.11)
OC	1 µg/m ³	1.02 (0.98 - 1.07)	1.01 (0.97 - 1.04)
Seasalt	1 µg/m ³	0.97 (0.97 - 0.98)	0.97 (0.96 - 0.98)

Abbreviations: HR, hazard ratio; 95% CI, 95% confidence interval.

Model description: Conditional logistic regression analyses of the effect of average 12 months air pollutant concentrations on asthma. Adjustment for age, calendar year, parental asthma, parental education and parental income (Model 3).

Table E13 Adjusted effects of air pollution on the risk of asthma and persistent wheezing by parental asthma.

Air pollutants	Increase	Parental asthma			
		HR (95% CI) No parents	HR (95% CI) Father	HR (95% CI) Mother	HR (95% CI) Both parents
O ₃	10 µg/m ³	0.96 (0.95 - 0.98)	0.97 (0.92 - 1.00)	0.95 (0.92 - 0.97)	0.96 (0.87 - 1.06)
NO ₂	10 µg/m ³	1.03 (1.02 - 1.05)	1.03 (0.99 - 1.08)	1.04 (1.01 - 1.06)	1.13 (1.03 - 1.24)
NO _x	10 µg/m ³	1.03 (1.02 - 1.04)	1.04 (1.01 - 1.07)	1.04 (1.02 - 1.06)	1.07 (1.01 - 1.14)
SO ₂	5 µg/m ³	1.01 (0.98 - 1.04)	1.06 (1.00 - 1.11)	1.03 (0.99 - 1.07)	1.12 (0.98 - 1.27)
PM _{2.5}	5 µg/m ³	1.04 (1.03 - 1.07)	1.09 (1.03 - 1.16)	1.05 (1.01 - 1.10)	1.10 (0.95 - 1.28)
PM ₁₀	5 µg/m ³	1.04 (1.02 - 1.07)	0.98 (0.91 - 1.05)	1.05 (1.00 - 1.10)	0.90 (0.76 - 1.08)
SIA	5 µg/m ³	1.02 (1.00 - 1.05)	1.02 (0.97 - 1.08)	0.98 (0.94 - 1.01)	0.97 (0.86 - 1.11)
SO ₄ ⁻	5 µg/m ³	1.00 (0.96 - 1.04)	1.08 (0.97 - 1.19)	1.01 (0.94 - 1.08)	0.88 (0.69 - 1.12)
NO ₃	5 µg/m ³	1.04 (1.02 - 1.07)	1.08 (1.00 - 1.15)	1.02 (0.97 - 1.07)	0.96 (0.83 - 1.12)
NH ₄ ⁺	1 µg/m ³	1.10 (1.02 - 1.19)	1.19 (0.97 - 1.47)	0.93 (0.81 - 1.07)	0.87 (0.53 - 1.42)
EC	1 µg/m ³	1.00 (0.93 - 1.07)	1.25 (1.05 - 1.49)	1.16 (1.03 - 1.31)	1.51 (0.98 - 2.32)
OC	1 µg/m ³	0.99 (0.95 - 1.03)	1.11 (1.02 - 1.22)	1.07 (1.01 - 1.14)	1.23 (0.98 - 1.54)
Seasalt	1 µg/m ³	0.97 (0.96 - 0.98)	0.96 (0.93 - 0.98)	0.97 (0.95 - 0.99)	0.91 (0.85 - 0.97)

Abbreviations: HR, hazard ratio; 95% CI, 95% confidence interval.

Model description: Conditional logistic regression analyses of the effect of average 12 months air pollutant concentrations on asthma. Adjustment for age, sex, calendar year, parental asthma, parental income and education (Model 3).

Table E14 Adjusted effects of air pollution on the risk of asthma and persistent wheezing by parental educational level.

Air pollutants	Increase	Parental educational level						
		HR (95% CI) Level 1	HR (95% CI) Level 2	HR (95% CI) Level 3	HR (95% CI) Level 4	HR (95% CI) Level 5	HR (95% CI) Level 6	HR (95% CI) Level 7
O ₃	10 µg/m ³	0.94 (0.89 - 0.99)	0.93 (0.91 - 0.96)	0.95 (0.93 - 0.97)	0.98 (0.95 - 1.02)	1.30 (0.99 - 1.08)	1.01 (0.94 - 1.09)	0.97 (0.88 - 1.07)
NO ₂	10 µg/m ³	1.04 (0.98 - 1.10)	1.04 (1.01 - 1.08)	1.05 (1.03 - 1.08)	1.01 (0.98 - 1.04)	0.98 (0.94 - 1.02)	0.94 (0.88 - 1.02)	0.99 (0.91 - 1.09)
NO _x	10 µg/m ³	1.02 (0.98 - 1.06)	1.04 (1.02 - 1.07)	1.04 (1.02 - 1.05)	1.03 (1.01 - 1.06)	0.98 (0.95 - 1.01)	0.97 (0.92 - 1.02)	1.00 (0.93 - 1.07)
SO ₂	5 µg/m ³	1.00 (0.93 - 1.08)	1.04 (1.00 - 1.09)	1.05 (1.02 - 1.08)	1.02 (0.98 - 1.06)	0.93 (0.88 - 0.98)	0.93 (0.85 - 1.02)	0.91 (0.81 - 1.01)
PM _{2.5}	5 µg/m ³	1.06 (0.97 - 1.15)	1.04 (0.99 - 1.09)	1.06 (1.03 - 1.09)	1.04 (0.99 - 1.10)	0.95 (0.89 - 1.03)	0.96 (0.85 - 1.09)	1.03 (0.89 - 1.21)
PM ₁₀	5 µg/m ³	1.03 (0.93 - 1.14)	1.04 (0.99 - 1.10)	0.99 (0.96 - 1.03)	1.02 (0.97 - 1.08)	1.07 (1.00 - 1.15)	1.18 (1.04 - 1.34)	1.08 (0.94 - 1.24)
SIA	5 µg/m ³	0.97 (0.90 - 1.04)	1.02 (0.98 - 1.06)	1.02 (0.99 - 1.04)	1.01 (0.97 - 1.05)	0.98 (0.92 - 1.04)	0.98 (0.88 - 1.11)	0.93 (0.81 - 1.08)
SO ₄ ⁻	5 µg/m ³	1.00 (0.88 - 1.15)	1.00 (0.93 - 1.09)	1.03 (0.98 - 1.09)	0.99 (0.92 - 1.07)	0.86 (0.76 - 0.96)	1.05 (0.86 - 1.29)	0.95 (0.72 - 1.24)
NO ₃	5 µg/m ³	0.95 (0.88 - 1.04)	1.03 (0.98 - 1.08)	1.05 (1.01 - 1.08)	1.04 (0.98 - 1.09)	0.99 (0.92 - 1.07)	1.08 (0.94 - 1.25)	1.07 (0.89 - 1.29)
NH ₄ ⁺	1 µg/m ³	0.93 (0.72 - 1.22)	1.07 (0.92 - 1.24)	1.04 (0.94 - 1.15)	1.01 (0.86 - 1.19)	1.04 (0.81 - 1.32)	1.24 (0.78 - 1.98)	0.67 (0.83 - 1.41)
EC	1 µg/m ³	1.08 (0.83 - 1.41)	1.25 (1.09 - 1.44)	1.23 (1.12 - 1.35)	0.96 (0.84 - 1.10)	0.70 (0.58 - 0.83)	0.72 (0.53 - 0.97)	0.59 (0.41 - 0.83)
OC	1 µg/m ³	1.03 (0.90 - 1.18)	1.11 (1.04 - 1.19)	1.09 (1.04 - 1.15)	0.97 (0.91 - 1.04)	0.83 (0.76 - 0.91)	0.83 (0.72 - 0.97)	0.75 (0.62 - 0.91)
Seasalt	1 µg/m ³	0.96 (0.93 - 1.00)	0.95 (0.94 - 0.97)	0.95 (0.94 - 0.97)	0.98 (0.97 - 1.00)	1.02 (0.99 - 1.05)	1.07 (1.00 - 1.12)	1.08 (1.01 - 1.16)

Abbreviations: HR, hazard ratio; 95% CI, 95% confidence interval.

Model description: Conditional logistic regression analyses of the effect of average 12 months air pollutant concentrations on asthma. Adjustment for age, sex, calendar, parental asthma and parental income. Parental education was generated adding the level of education (level 1-4) for each parent generating a 7-level categorical variable for parental education.

Table E15 Adjusted effects of air pollution on the risk of asthma and persistent wheezing by parental income.

Air pollutant		Parental income level								
		HR (95% CI)	HR (95% CI)	HR (95% CI)	HR (95% CI)	HR (95% CI)	HR (95% CI)	HR (95% CI)	HR (95% CI)	HR (95% CI)
		Level 1	Level 2	Level 3	Level 4	Level 5	Level 6	Level 7	Level 8	Level 9
Increase										
O ₃	10 µg/m ³	1.09 (0.99 - 1.21)	0.99 (0.93 - 1.07)	0.96 (0.92 - 1.01)	1.01 (0.98 - 1.05)	0.98 (0.96 - 1.01)	0.96 (0.94 - 0.98)	0.96 (0.93 - 0.99)	0.97 (0.92 - 1.02)	1.04 (0.95 - 1.12)
NO ₂	10 µg/m ³	0.76 (0.69 - 0.84)	0.89 (0.83 - 0.96)	1.00 (0.95 - 1.05)	0.97 (0.93 - 1.01)	1.01 (0.98 - 1.04)	1.04 (1.01 - 1.07)	1.05 (1.02 - 1.09)	1.06 (1.01 - 1.11)	0.99 (0.91 - 1.09)
NO _x	10 µg/m ³	0.88 (0.82 - 0.94)	0.94 (0.89 - 1.00)	1.00 (0.96 - 1.04)	1.00 (0.97 - 1.03)	1.00 (0.98 - 1.02)	1.04 (1.02 - 1.06)	1.04 (1.02 - 1.07)	1.03 (0.99 - 1.06)	1.01 (0.96 - 1.07)
SO ₂	5 µg/m ³	0.82 (0.74 - 0.91)	0.95 (0.88 - 1.04)	1.00 (0.95 - 1.07)	0.96 (0.92 - 1.01)	0.99 (0.95 - 1.03)	1.02 (0.98 - 1.05)	1.05 (0.99 - 1.09)	1.07 (0.99 - 1.15)	1.03 (0.92 - 1.16)
PM _{2.5}	5 µg/m ³	1.18 (0.97 - 1.43)	1.11 (0.97 - 1.28)	1.16 (1.06 - 1.27)	1.09 (1.03 - 1.17)	1.05 (1.01 - 1.09)	1.04 (1.00 - 1.08)	1.03 (0.98 - 1.08)	0.95 (0.88 - 1.03)	0.99 (0.87 - 1.14)
PM ₁₀	5 µg/m ³	1.20 (1.05 - 1.37)	1.23 (1.10 - 1.39)	1.12 (1.03 - 1.22)	1.07 (1.00 - 1.14)	1.07 (1.01 - 1.12)	0.99 (0.94 - 1.04)	1.04 (0.98 - 1.11)	0.95 (0.86 - 1.05)	0.98 (0.84 - 1.16)
SIA	5 µg/m ³	1.01 (0.89 - 1.15)	1.02 (0.96 - 1.12)	1.02 (0.95 - 1.08)	1.04 (0.99 - 1.09)	1.00 (0.97 - 1.04)	1.01 (0.97 - 1.04)	1.01 (0.97 - 1.05)	1.02 (0.94 - 1.09)	1.01 (0.89 - 1.15)
SO ₄ ⁻	5 µg/m ³	0.86 (0.59 - 1.23)	0.88 (0.70 - 1.11)	0.97 (0.84 - 1.11)	0.97 (0.89 - 1.07)	0.95 (0.89 - 1.02)	1.03 (0.96 - 1.09)	0.98 (0.89 - 1.06)	1.03 (0.89 - 1.19)	0.95 (0.74 - 1.21)
NO ₃	5 µg/m ³	0.96 (0.79 - 1.17)	1.04 (0.89 - 1.21)	1.18 (1.07 - 1.28)	1.05 (0.98 - 1.12)	1.05 (1.00 - 1.10)	1.00 (0.96 - 1.04)	1.07 (1.02 - 1.12)	1.06 (0.98 - 1.15)	1.22 (1.06 - 1.40)
NH ₄ ⁺	1 µg/m ³	1.83 (1.09 - 3.06)	1.75 (1.16 - 2.62)	1.71 (1.32 - 2.22)	1.19 (0.99 - 1.44)	1.11 (0.96 - 1.28)	0.98 (0.86 - 1.12)	1.01 (0.85 - 1.19)	0.88 (0.65 - 1.18)	1.31 (0.77 - 2.23)
EC	1 µg/m ³	0.28 (0.18 - 0.42)	0.48 (0.34 - 1.66)	0.76 (0.62 - 0.96)	0.75 (0.63 - 0.89)	0.85 (0.75 - 0.97)	1.13 (1.00 - 1.27)	1.14 (0.99 - 1.32)	1.14 (0.92 - 1.43)	0.77 (0.54 - 1.08)
OC	1 µg/m ³	0.54 (0.43 - 0.65)	0.70 (0.59 - 0.83)	0.87 (0.78 - 0.98)	0.86 (0.79 - 0.94)	0.92 (0.85 - 0.97)	1.05 (0.99 - 1.12)	1.06 (0.98 - 1.13)	1.06 (0.94 - 1.18)	0.86 (0.72 - 1.03)
Seasalt	1 µg/m ³	1.11 (1.05 - 1.18)	1.07 (1.02 - 1.12)	1.00 (0.97 - 1.03)	1.01 (0.98 - 1.03)	0.99 (0.97 - 1.01)	0.96 (0.95 - 0.98)	0.95 (0.93 - 0.98)	0.95 (0.92 - 0.98)	0.95 (0.89 - 1.02)

Abbreviations: HR, hazard ratio; 95% CI, 95% confidence interval.

Model description: Conditional logistic regression analyses of the effect of average 12 months air pollutant concentrations on asthma. Adjustment for age, sex, calendar year, parental asthma and parental education. Parental income was generated adding the level of income (level 1-5) for each parent generating a 9-level variable for parental income.

Table E16 Adjusted effects of air pollution on the risk of asthma and persistent wheezing by maternal smoking during pregnancy.

Air pollutants	Increase	Smoking status		
		HR (95% CI)	HR (95% CI)	HR (95% CI)
		Non-smoker	Quit smoking during pregnancy	Smoking during pregnancy
O ₃	10 µg/m ³	0.97 (0.96 - 0.98)	0.98 (0.84 - 1.15)	0.94 (0.91 - 0.97)
NO ₂	10 µg/m ³	1.02 (1.01 - 1.04)	1.01 (0.86 - 1.18)	1.07 (1.04 - 1.11)
NO _x	10 µg/m ³	1.02 (1.02 - 1.04)	1.06 (0.95 - 1.19)	1.05 (1.03 - 1.07)
SO ₂	5 µg/m ³	1.00 (0.98 - 1.02)	1.12 (0.91 - 1.37)	1.07 (1.02 - 1.12)
PM _{2.5}	5 µg/m ³	1.05 (1.03 - 1.08)	1.03 (0.81 - 1.32)	1.07 (1.02 - 1.13)
PM ₁₀	5 µg/m ³	1.05 (1.02 - 1.08)	0.92 (0.72 - 1.18)	1.00 (0.94 - 1.07)
SIA	5 µg/m ³	1.00 (0.99 - 1.03)	0.91 (0.73 - 1.12)	0.99 (0.95 - 1.03)
SO ₄ ⁻	5 µg/m ³	0.99 (0.96 - 1.03)	0.84 (0.56 - 1.25)	0.99 (0.92 - 1.07)
NO ₃	5 µg/m ³	1.03 (1.01 - 1.06)	0.94 (0.71 - 1.24)	1.04 (0.99 - 1.09)
NH ₄ ⁺	1 µg/m ³	1.07 (0.99 - 1.15)	0.48 (0.21 - 1.08)	1.03 (0.88 - 1.21)
EC	1 µg/m ³	0.95 (0.90 - 1.02)	1.70 (0.82 - 3.51)	1.26 (1.09 - 1.47)
OC	1 µg/m ³	0.97 (0.93 - 1.00)	1.33 (0.92 - 1.95)	1.12 (1.04 - 1.21)
Seasalt	1 µg/m ³	0.97 (0.97 - 0.98)	0.92 (0.82 - 1.02)	0.96 (0.94 - 0.98)

Abbreviations: HR, hazard ratio; 95% CI, 95% confidence interval.

Model description: Conditional logistic regression analyses of the effect of average 12 months air pollutant concentrations on asthma. Adjustment for age, sex, calendar year, parental asthma, parental education and parental income.

Table E17 Adjusted effects of air pollution on the risk of asthma and persistent wheezing by age groups.

Air pollutants	Increase	Age groups	
		HR (95% CI)	HR (95% CI)
		<6 years	≥6 years
O ₃	10 µg/m ³	0.96 (0.96 - 0.97)	0.89 (0.85 - 0.93)
NO ₂	10 µg/m ³	1.03 (1.02 - 1.04)	1.16 (1.11 - 1.22)
NO _x	10 µg/m ³	1.03 (1.03 - 1.04)	1.11 (1.07 - 1.15)
SO ₂	5 µg/m ³	1.01 (1.00 - 1.04)	1.11 (1.05 - 1.18)
PM _{2.5}	5 µg/m ³	1.05 (1.03 - 1.07)	1.08 (0.99 - 1.16)
PM ₁₀	5 µg/m ³	1.05 (1.02 - 1.07)	0.93 (0.86 - 1.00)
SIA	5 µg/m ³	1.01 (0.99 - 1.02)	0.98 (0.92 - 1.05)
SO ₄ ⁻	5 µg/m ³	1.01 (0.97 - 1.03)	0.93 (0.81 - 1.08)
NO ₃	5 µg/m ³	1.03 (1.02 - 1.05)	0.97 (0.87 - 1.07)
NH ₄ ⁺	1 µg/m ³	1.05 (0.98 - 1.11)	0.72 (0.54 - 0.94)
EC	1 µg/m ³	1.02 (0.97 - 1.08)	2.03 (1.63 - 2.51)
OC	1 µg/m ³	1.00 (0.97 - 1.02)	1.43 (1.29 - 1.59)
Seasalt	1 µg/m ³	0.97 (0.97 - 0.98)	0.92 (0.89 - 0.95)

Abbreviations: HR, hazard ratio; 95% CI, 95% confidence interval

Model description: Conditional logistic regression analyses of the effect of average 12 months air pollutant concentrations on asthma. Adjustment for sex, calendar year, parental asthma, parental income and parental education (Model 3).